

— CONTAINERISED DATA CENTER

Kelvin Flexbox™

An ≈1 MW containerised compute unit, purpose-built to mitigate curtailment and dispatch on demand. Power in. Compute. Heat out.

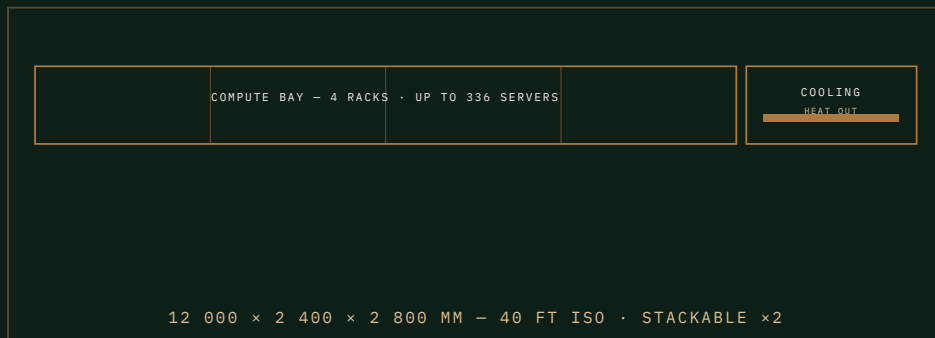
≈1 MW

PER 40 FT ISO — TURNING WASTE ENERGY INTO HIGH-PERFORMANCE COMPUTE. STACKABLE ×2 ≈ 2 MW.



VISUALIZATION KELVIN FLEXBOX™ — 40 FT ISO

- 01 **Compute**
High-density load whenever the grid does not call.
- 02 **Flexibility**
Sub-0.7 s dispatch — FCR-D · aFRR · mFRR.
- 03 **Heat as an asset**
Capture designed in — district, greenhouse, process.



VISUALIZATION STACKABLE ×2 ≈ 2 MW

≈1 MW

OPERATING POWER PER CONTAINER

40 ft

STANDARD ISO FOOTPRINT

<0.7 s

DISPATCH RESPONSE TIME

01 SYSTEM OVERVIEW

A containerised megawatt that earns three ways

The Kelvin Flexbox packages high-density compute servers, smart PDUs, forced-air or liquid cooling and networking into a standard 40 ft ISO container (20 ft on request). It connects at low or medium voltage and is built to start, pause and resume in well under a second — turning a megawatt of compute into a controllable, market-facing asset.

Because the unit runs compute whenever it is not dispatched, every hour of operation produces compute revenue, while flexibility and ancillary-services participation add a second, uncorrelated stream. Kelvin's own software handles dispatch, control and monitoring in real time across every unit. Each unit is paid on standby, paid on activation and paid in compute. Heat capture is designed in from day one.

SUB-0.7 S RESPONSE	INSTANT PAUSE	UNLIMITED CURTAILMENT	LV / MV	STACKABLE ×2
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— REVENUE STACK

01

Compute

Compute revenue as a continuous baseline whenever the unit is running — uncorrelated with the power market.

02

Flexibility (FaaS)

Balancing & ancillary services — sub-second response, with no cycling penalty for pausing and resuming. Absorbs surplus and curtailed generation.

03

Heat as an asset

Every kilowatt of compute leaves as heat. Capture is designed in from day one — district networks, greenhouses, process heat.

02 APPLICATIONS

ONE ASSET ACROSS THE FLEXIBILITY STACK

→ Behind-the-meter	Co-locate at generation or substation
→ Frequency reserves	FFR · FCR-D · aFRR · mFRR — markets the load is built for
→ Demand response	Curtail on signal, resume instantly
→ Compute revenue	Continuous baseline, uncorrelated with power
→ Renewables off-take	Absorb surplus and curtailed generation
→ Grid balancing	Fast, controllable flexible load

03 FLEXIBILITY, COMPARED

Compute flexibility compared to batteries

Battery energy storage is a well-established flexibility solution. Compute-based flexibility offers a different economic and operational profile — especially where **compute, flexibility and heat** sit on the same megawatt, and **continuous dispatch** matters. Grid infrastructure doesn't have a wires problem — it has a *flexibility problem*.

	BATTERY STORAGE (BESS)	COMPUTE FLEXIBILITY
UPFRONT COST	— €200–500k per MWh of capacity	✓ Near-zero marginal cost — servers already deployed
CYCLE DEGRADATION	— Capacity reduces with every activation	✓ Zero degradation — servers dispatch without wearing out
DURATION LIMIT	— Fixed energy reservoir; duration is capped	✓ Unlimited — servers run continuously, around the clock
RECOVERY TIME	— Downtime required between activations	✓ None — assets available immediately after each dispatch
RISK PROFILE	— Fire and thermal-runaway risk with chemical storage	✓ No chemical storage, no associated hazards
REVENUE STREAMS	— Grid services only	✓ Compute + grid services + heat, on the same megawatt

Batteries store energy. **Compute earns while it flexes — and heat is designed in.**

BESS FIGURES ARE INDICATIVE INDUSTRY RANGES · THE TWO TECHNOLOGIES CAN BE COMPLEMENTARY WITHIN A FLEXIBILITY PORTFOLIO

04 CONFIGURATION & CAPACITY

Power, configuration and layout

CONFIGURATION	SERVERS / UNIT	OPERATING POWER	CONNECTION	FOOTPRINT
Kelvin Flexbox™	up to 336	≈ 1 MW	LV or MV	40 ft · 1 layer
Stacked pair	up to 672	≈ 2 MW	LV or MV	40 ft · 2 layers

MV TRANSFORMER AND SWITCHGEAR SUPPLIED ON REQUEST · OPERATING POWER TRACKS LOADED SERVER DRAW (= 1 MW PER CONTAINER) · 20 FT VARIANT ON REQUEST – FIGURES REFER TO THE 40 FT UNIT

05 ANATOMY

Four racks, heat loop, 40 ft ISO



VISUALIZATION INTERNALS — 4 RACKS · UP TO 336 SERVERS · HEAT LOOP



VISUALIZATION DOORS, LOUVERS, COPPER K

GRID (LV OR MV) → TRANSFORMER → MV/LV SWITCHGEAR → FLEXBOX (2 × MCCB 1250 A · SMART PDU · 4 RACKS · AIR OR LIQUID · HEAT FLANGE). KELVIN SOFTWARE: CONTROL · DISPATCH · MONITORING.

06 DEPLOYMENT

Grid-connected

Stand-alone at the substation with its own transformer and switchgear — connects at LV or MV as a controllable, market-facing load.

Behind-the-meter

Co-located at wind, solar or hydro to absorb curtailed or low-price generation on site.

Stacked cluster

Units stack up to two layers and scale out to multi-megawatt sites on a single connection.

07 TECHNICAL DATA

Specifications

Kelvin Flexbox™ · ≈1 MW · air- or liquid-cooled 40 ft ISO.

GENERAL	
Container type	Standard 40 ft (20 ft on request)
External dimensions (L×W×H)	12 000 × 2 400 × 2 800 mm
Gross weight	≈ 10 t
Vertical stacking	Up to 2 layers · ≈ 2 MW
Heat capture	Designed in · district / greenhouse / process
Grid connection	LV or MV

ENVIRONMENT	
Operating temperature	−25 to +40 °C
Operating humidity	10–90% RH
Storage temperature	−35 to +70 °C
Storage humidity	5–95% RH
Altitude	≤ 2000 m
Cooling	Forced air or liquid

ELECTRICAL & POWER	
Operating power	≈ 1 MW (1,000 kW)
Input voltage	3P/N/PE AC 380–415 V
Frequency	50/60 Hz
Connection point	LV distribution cabinet
Main protection	2 × MCCB 1250 A
Rated power per server	4 kW

BALANCE OF PLANT (SITE)	
MV transformer	MV → 380–415 V
MV/LV switchgear	Protection · isolation · metering
LV distribution	To container cabinet
Earthing	Per local code
Required grid capacity	Sized to rated load / unit
Standing area	Cleared access per side

COOLING	
Type	Forced-air, negative-pressure · liquid option
Number of fans	16
Fan dimensions	1,220 × 1,220 × 385 mm
Air volume per fan	20,000–30,000 m³/h
Fan supply	415 V
Drive / rating	Belt · IP58

POWER DISTRIBUTION	
Distribution cabinet	800 × 600 × 2,000 mm · IP51
Main breakers	2 × MCCB 1250 A
Branch breakers	14 × MCB 100 A
PDU dimensions	2,000 × 49 × 100 mm · IP51
PDU input	380/480 V AC 3P+N+PE
PDU max. current	100 A
PDU outlets	1×C13 + 12×P33 IEC

COMPUTE RACKS & CAPACITY	
Racks per container	4
Layers per rack	7 · 12 servers/layer
Rack dimensions	2,444 × 2,560 × 350 mm
Servers per rack	up to 84
Servers per container	up to 336

CONTROL & MONITORING	
Dispatch response	< 0.7 s
Control software	Kelvin · in-house
Monitoring	Web & app · real-time
Sensors	Temperature & humidity
Temp. measuring range	−40 to +85 °C · ext. probe
Accuracy	±0.3 °C · ±3% RH
Network	10/100 Base-T · RJ45
Interface / protocols	Ethernet · Modbus TCP opt.

TECHNICAL DATA BASED ON THE KELVIN FLEXBOX™ WORKING SPECIFICATION. OPERATING POWER REFLECTS TYPICAL LOADED SERVER DRAW. SITE-SPECIFIC VALUES TO BE CONFIRMED DURING SITE ENGINEERING.

COMPANY

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